

1(a)	blood contained in (blood) vessels AW or blood contained in <i>any three of</i> heart, arteries, veins, capillaries ; systemic and pulmonary, systems / circulation ; A described <i>if circulations not named</i> e.g. for each complete circuit (round the body) blood passes through heart twice blood transported from heart to lungs and back, then to (rest of) body and back	2					
1(b)	<table border="1"><tr><td>2</td></tr><tr><td>4</td></tr><tr><td>1</td></tr><tr><td>3</td></tr><tr><td>5</td></tr></table> 1st and 5th boxes (2 and 5) correct ; 2nd and 4th boxes (4 and 3) correct ;	2	4	1	3	5	2
2							
4							
1							
3							
5							
1(c)(i)	<i>assume answer refers to arteries unless stated otherwise</i> withstand / AW, higher pressure (of blood) ; prevent rupturing / bursting (from high blood pressure) ; I collapsing <i>one from</i> thicker / AW, tunica media ; more elastic, tissue / fibres, and (smooth) muscle tissue ; more / AW, elastic, tissue / fibres, to maintain, blood pressure / blood flow ; more (smooth) muscle to maintain, blood pressure / blood flow ;	max 2					

Question	Answer	Marks
1(c)(ii)	<i>carbon monoxide max 3</i> forms carboxyhaemoglobin ; A binds to haemoglobin less haemoglobin available to bind oxygen / haemoglobin has greater affinity for carbon monoxide / AW ; reduces, percentage saturation of haemoglobin (with oxygen) / AW A less oxygen binds to haemoglobin I prevents oxygen binding or reduces oxygen carrying capacity of, haemoglobin / blood ; <i>nicotine max 3</i> increases heart rate ; increases blood pressure ; constricts, arteries / arterioles ; A vasoconstriction makes platelets sticky / promotes clotting / promotes thrombosis ; <i>for either</i> damages, endothelium / tunica intima / lining of blood vessels ;	max 4
	Total:	10